

SIMON MUTHUNGU

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Summary

Results-driven AI Engineer & Full-Stack Developer with 8 years Python (5 in production). Built scalable APIs, RAG systems, and predictive models. Expert in Django/FastAPI, LangChain, AWS, Docker/K8s. Proven leadership delivering production systems on time. Seeking a mid-level role focused on creating impactful and efficient software solutions driving cutting-edge, ethically-aligned solutions for complex business challenges.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, SQL, PHP, Elixir
- **Backend Frameworks:** Django, Flask, FastAPI, Elixir Phoenix, Laravel
- **Frontend Frameworks:** ReactJS, NextJS, Tailwind CSS, JQuery
- **AI/LLM Development:** Artificial Intelligence, Machine Learning, Generative AI, LLM Application Development, AI Agent Engineering & Architecture
- **Agent Frameworks:** Lang Chain, LangGraph, Vercel AI SDK, Hugging Face JS Libraries, n8n automation
- **Cloud/MLOps:** Docker & Kubernetes (K8s), **AWS ECS & Fargate**, **AWS ECR**, **Application Load Balancers (ALB)**, **VPC Networking**, **AWS CloudWatch**, EC2, ECS, ECR, **Terraform**, Git, Jira, CI/CD Pipelines
- **Databases:** MySQL, PostgreSQL, Supabase, Apache Spark, Vector Databases (Pinecone & Weaviate for RAG)
- **Tools:** Git, Docker, Jira, Confluence
- **Methodologies:** Agile, Scrum, DevOps
- **Advanced AI/ML:** **Reinforcement Learning** (for adaptive agents), **Neural Network Architectures** (Transformer Models), **AI Security & Governance** (Prompt Injection Mitigation, Model Armor), **Explainable AI**

- Leadership

Professional Experience

Agent Complynt – AI Compliance Analyst

Dec 2025

Designed and engineered **Agent Complynt**, a RAG-powered compliance intelligence system for SMEs and NGOs. Built a full ingestion + retrieval pipeline (PDF parsing, chunking, batching, embedding, vector indexing) using **FastAPI**, **Pinecone**, **LangGraph**, and **Next.js**, enabling automated analysis of tender documents, compliance certificates, and legal acts. Implemented a streaming LangGraph workflow for real-time regulatory reasoning, anomaly detection, and vendor vetting. Delivered a production-ready demo showcasing AI-driven compliance automation aimed at improving SME readiness, reducing risk, and strengthening national business resilience.

LIVE RAG AGENT LINK: <https://agent-complynt-v2.vercel.app>

AURA Agent – Serverless LLM Deployment

Nov 2025

Architected and deployed AURA Agent, a production-scale conversational AI system built with LangServe & LangChain, and deployed on Vercel serverless functions, and a Supabase PostgreSQL backend. Designed a robust, low-latency service layer enabling reliable, scalable agent interactions without dedicated servers. Implemented secure data access, optimized cold-start performance, and ensured high-availability inference at scale.

Live Agent:

<https://agent-aura.vercel.app/aura-agent/playground/>

Developed **workflow automation** using **Zapier, Make, and n8n**, improving operational speed and efficiency by 70% across integrated systems.

Software Engineer & Project Lead, Home Dispo (homedispo.com)

February 2025 - June 2025

- **Architected and deployed** a suite of APIs that reduced data retrieval time by 30% for the US-based startup.
- **Led** a cross-functional team of 4 to deliver a real estate platform, accelerating feature delivery time by 20% by implementing agile workflows.
- **Optimized** backend services and database queries to enhance application performance, and **tuned** performance metrics to ensure optimal application response times.
- **Led QA efforts** and used tools like **Playwright** to run automated tests, catching bugs early and ensuring a stable release cycle.
- **Developed** internal tools that streamlined operations and automated manual tasks increasing efficiency and productivity by 60% to support user experience goals.

AI & ML Lead Engineer, Proacted & Enactus Club (enactusmasenochapter.africa)

March 2023 - June 2025

- **Designed and implemented predictive** machine learning models that improved operational efficiency by 25% through automation for proacted students.
- **Developed** a predictive analytics model that processed large datasets to guide data-driven business decisions.
- **Led** cross-functional teams to successfully deliver multiple AI-powered projects on time and within scope at Enactus.

- **Refactored** core codebase, improving execution speed and reducing resource consumption.

Production Team Leader, Househub (househub.co.ke)

September 2023 - January 2025

- **Built** core backend functionality and API integrations that ensured smooth property syncing and fast search performance.
- **Drove** product strategy and execution for a property discovery platform, overseeing design and data workflows to deliver user-first features.
- **Guided** a lean team from concept to launch with strong collaboration and technical leadership

AI Lead Developer, Proacted

March 2023 - January 2025

- **Designed** innovative AI algorithms and trained predictive models that enhanced efficiency and accuracy for high school students.
- **Developed** robust software solutions that improved operational workflows and user experience.
- **Created and optimized** machine learning models for predictive analytics and data-driven decision-making.
- **Led** cross-functional teams to successfully deliver projects on time and within scope.

EDUCATION

Maseno University (2022)

Bachelor of Science - BS, Computer Science

- Learnt computer science fundamentals
- Learnt core foundational computing principles.

- Learnt modern frameworks of machine learning and artificial intelligence

CERTIFICATION AND LICENSES

Google Developer Student Club (GDGoC) AI/ML Lead - 2025

Nexus Backend AI Developer - 2024

Organiser Deep Learning IndabaX - 2024

MUCISA (Maseno University Computing & Informatics Student Association)

Hackathon Lead Organiser - 2024

Kaps Inter-University competition - Top Team 2024